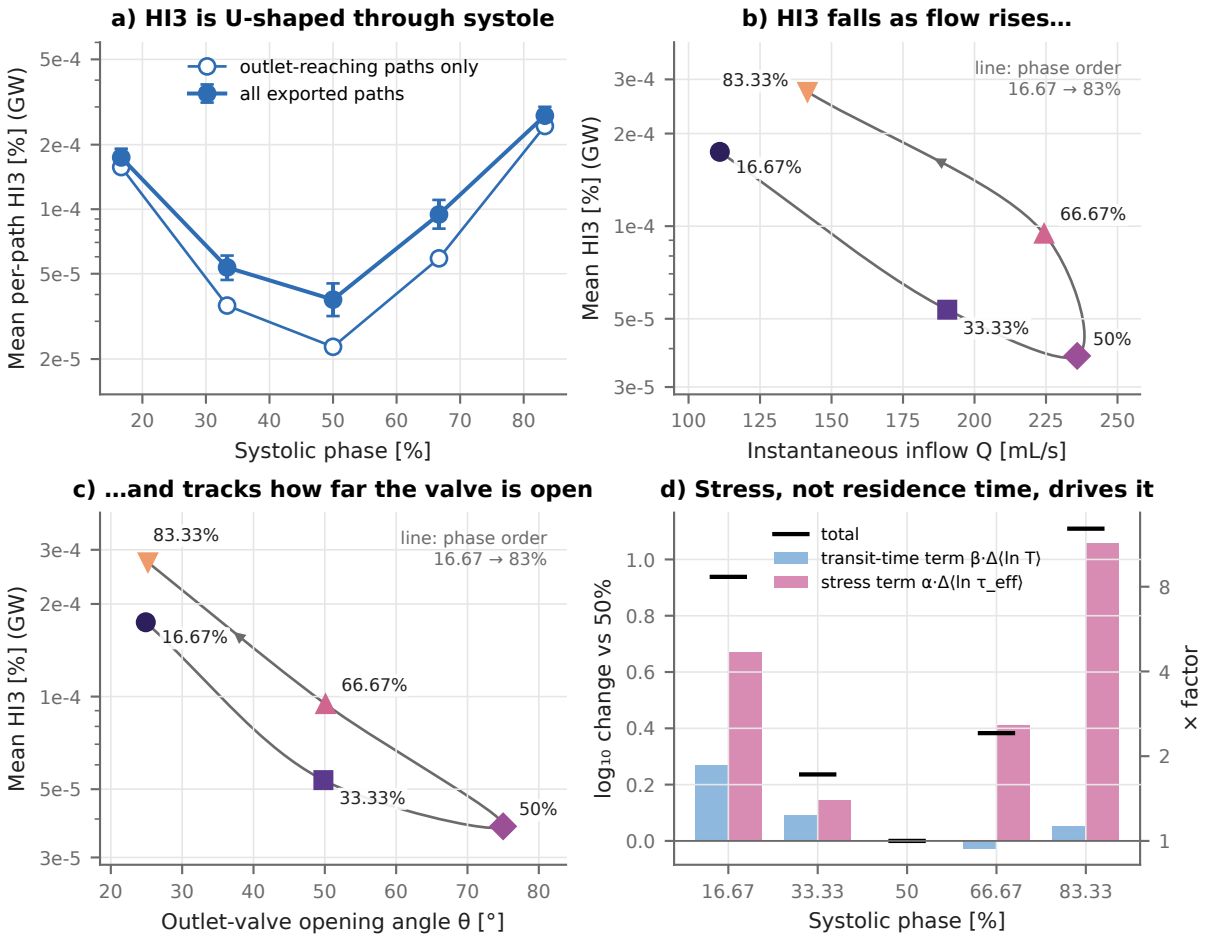
a) HI3 is U-shaped through systole
outlet-reaching paths only
all exported paths
Mean per-path HI3 [%] (GW)
5e-4
2e-4
1e-4
5e-5
2e-5
20
30
40
50
60
70
80
Systolic phase [%]
b) HI3 falls as flow rises…
line: phase order
16.67 → 83%
83.33%
16.67%
66.67%
33.33%
50%
Mean HI3 [%] (GW)
3e-4
2e-4
1e-4
5e-5
3e-5
100
125
150
175
200
225
250
Instantaneous inflow Q [mL/s]
c) …and tracks how far the valve is open
line: phase order
16.67 → 83%
83.33%
16.67%
66.67%
33.33%
50%
Mean HI3 [%] (GW)
3e-4
2e-4
1e-4
5e-5
3e-5
20
30
40
50
60
70
80
Outlet-valve opening angle θ [°]
d) Stress, not residence time, drives it
total
transit-time term β·Δ⟨ln T⟩
stress term α·Δ⟨ln τ_eff⟩
log₁₀ change vs 50%
1.0
0.8
0.6
0.4
0.2
0.0
× factor
8
4
2
1
16.67
33.33
50
66.67
83.33
Systolic phase [%]